SALES ANALYTICS

StyleSense is a growing clothing retail company that sells Men's, Women's, and Kids' clothing across Physical stores and Online channels. Like many retail businesses today, StyleSense faces growing competition from large e-commerce platforms and needs to make smarter, data-driven decisions to stay profitable.

Business Problem

Despite consistent sales, StyleSense's :

- Profit margins are unclear – cost of goods, payroll, rent, and marketing all eat into revenue.
- Seasonality is not being planned for – winter and summer products go out of stock or sit unsold.
- Promotions and discounts are used freely – but no one knows if they're actually generating profit.
- Customer segments are not being leveraged – VIP customers may be leaving due to lack of retention programs.
- Online vs physical performance is unknown – marketing spend may be mis-allocated.

Project Objective

This project analyzes 18 months (Jan 2024 – Jun 2025) of sales, customer, inventory, and expense data to:

- Understand what has been selling and who is buying (Descriptive).
- Diagnose why performance is the way it is (Diagnostic).
- Predict future revenue trends (Predictive).
- Recommend clear actions management can take to improve ROI (Prescriptive).

Data Understanding

- clothing_sales.csv(10,000 rows)-every transaction: product, channel, price, discount, profit.
- clothing_customers.csv(200 rows)-customer profiles: demographics, segments, purchase history.
- clothing_expenses.csv(1,300 rows)-business costs by category (rent, payroll, COGS, marketing).
- clothing_inventory.csv(13,600 rows)-daily stock levels per product with reorder triggers.
- clothing_products.csv(25 rows)-product master list with base price and cost.

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
from sklearn.linear_model import LinearRegression
```

```python
from sklearn.metrics import r2_score, mean_absolute_error
import warnings

warnings.filterwarnings('ignore')
```

In [2]:

```python
# Chart styling
sns.set_theme(style="whitegrid")
plt.rcParams['figure.dpi'] = 100
plt.rcParams['font.size'] = 10

BLUE   = '#3498DB'
GREEN  = '#27AE60'
RED    = '#E74C3C'
ORANGE = '#E67E22'
PURPLE = '#8E44AD'
GOLD   = '#F39C12'
TEAL   = '#1ABC9C'
GREY   = '#95A5A6'
```

In [3]:

```python
# Load all five datasets
sales     = pd.read_csv('clothing_sales.csv')
customers = pd.read_csv('clothing_customers.csv')
expenses  = pd.read_csv('clothing_expenses.csv')
inventory = pd.read_csv('clothing_inventory.csv')
products  = pd.read_csv('clothing_products.csv')
```

In [4]:

```python
print(f'  Sales rows:      {len(sales):,}')
print(f'  Customer rows:   {len(customers):,}')
print(f'  Expense rows:    {len(expenses):,}')
print(f'  Inventory rows:  {len(inventory):,}')
print(f'  Product rows:    {len(products):,}')
```

```
Sales rows:      10,039
Customer rows:   200
Expense rows:    1,322
Inventory rows:  13,675
Product rows:    25
```

In [5]:

```python
#Examine sales data structure
#Looking at the last few rows gives me a sense of: what columns exist,data types (dat
#and the most recent transactions (end of June 2025)
sales.tail()
```

Out[5]:

	sale_id	date	store_channel	payment_method	product_id	product_name	category
10034	S01581	2024-03-22	Physical	Credit Card	P010	Women's Blouse	Women
10035	S05467	2024-10-29	Physical	Debit Card	P021	Unisex Socks (3-pack)	All
10036	S06941	2025-01-11	Online	PayPal	P006	Men's Puffer Jacket	Men
10037	S06299	2024-12-17	Online	Debit Card	P021	Unisex Socks (3-pack)	All
10038	S00246	2024-01-07	Online	PayPal	P020	Unisex Cap	All

Data Cleaning_ Sales_csv

In [6]:

```
#Check for missing values
sales.isnull().sum()
```

Out[6]:

```
sale_id              0
date                 0
store_channel        0
payment_method       5
product_id           0
product_name         0
category             0
sub_category         0
quantity_sold        0
unit_price           0
discount_pct         0
discount_amount      0
total_sale           0
cost_total           0
gross_profit         0
customer_id        101
is_promotion         0
is_returned          0
dtype: int64
```

There are :

- 5 missing payment_methods
- 101 missing customer_ids

In [7]:

```
#check for nulls in sale_id column
sales.sale_id.isnull().sum()
```

Out[7]:

```
np.int64(0)
```

In [8]:

```
#check for duplicates in sale_id column
sales['sale_id'].duplicated().sum()
```

Out[8]: np.int64(50)

- There are 50 duplicated ids in sales_id column.

In [9]:

```
# See the duplicate rows
dupes = sales[sales['sale_id'].duplicated(keep=False)].sort_values('sale_id')
print(dupes[['sale_id', 'date', 'customer_id', 'product_id', 'quantity_sold', 'total_
```

	sale_id	date	customer_id	product_id	quantity_sold	total_sale
32	S00033	2024-01-02	C150	P020	1	11.99
9993	S00033	2024-01-02	C150	P020	1	11.99
145	S00146	2024-01-04	C047	P018	1	38.24
10020	S00146	2024-01-04	C047	P018	1	38.24
10038	S00246	2024-01-07	C111	P020	3	44.97
245	S00246	2024-01-07	C111	P020	3	44.97
248	S00249	2024-01-07	C028	P023	1	23.74
9989	S00249	2024-01-07	C028	P023	1	23.74
10031	S00273	2024-01-07	C200	P013	3	152.97
272	S00273	2024-01-07	C200	P013	3	152.97
816	S00817	2024-02-10	C179	P005	3	170.97
10004	S00817	2024-02-10	C179	P005	3	170.97
1033	S01034	2024-02-17	C045	P011	1	39.99
10010	S01034	2024-02-17	C045	P011	1	39.99
1157	S01158	2024-02-25	C141	P010	2	69.98
10011	S01158	2024-02-25	C141	P010	2	69.98
1472	S01473	2024-03-17	C184	P018	1	44.99
10000	S01473	2024-03-17	C184	P018	1	44.99
1580	S01581	2024-03-22	C124	P010	3	94.47
10034	S01581	2024-03-22	C124	P010	3	94.47

- The duplicates appear to be exact duplicates,suggesting a system glitch during data export. So i remove duplicates and I keep='first' to preserve one copy of each transaction. This reduces my dataset from 10,039 to 9,989 rows - a 50-row reduction.

In [10]:

```
# drop duplicates
sales = sales.drop_duplicates(subset='sale_id', keep='first')
print(sales.shape)
```

```
(9989, 18)
```

In [11]:

```
#check customer_id column
sales.customer_id
```

Out[11]:
```
0       C032
1       C151
2       C010
3       C017
4       C089
        ...
9984    C052
9985    C105
9986    C109
9987    C028
9988    C143
Name: customer_id, Length: 9989, dtype: str
```

In [12]:
```
# check for duplicates
sales.customer_id.duplicated().sum()
```

Out[12]:
```
np.int64(9788)
```

- The duplicated values in customer_id are likely to be because a customer can make purchases in different days, making them a repeat client.

In [13]:
```
# check column names
sales.columns
```

Out[13]:
```
Index(['sale_id', 'date', 'store_channel', 'payment_method', 'product_id',
       'product_name', 'category', 'sub_category', 'quantity_sold',
       'unit_price', 'discount_pct', 'discount_amount', 'total_sale',
       'cost_total', 'gross_profit', 'customer_id', 'is_promotion',
       'is_returned'],
      dtype='str')
```

In [14]:
```
# check for column data types
sales.dtypes
```

Out[14]:
```
sale_id                str
date                   str
store_channel          str
payment_method         str
product_id             str
product_name           str
category               str
sub_category           str
quantity_sold        int64
unit_price         float64
discount_pct         int64
discount_amount    float64
total_sale         float64
cost_total         float64
gross_profit       float64
customer_id            str
is_promotion          bool
is_returned           bool
dtype: object
```

In [15]:

```
# convert date column ino data data type
sales.date= pd.to_datetime(sales['date'])
```

In [16]:

```
# convert all string columns to object data type
str_cols = sales.select_dtypes(include='string').columns
sales[str_cols] = sales[str_cols].astype(object)
sales.dtypes
```

Out[16]:

```
sale_id                    object
date               datetime64[us]
store_channel              object
payment_method             object
product_id                 object
product_name               object
category                   object
sub_category               object
quantity_sold               int64
unit_price                float64
discount_pct                int64
discount_amount           float64
total_sale                float64
cost_total                float64
gross_profit              float64
customer_id                object
is_promotion                 bool
is_returned                  bool
dtype: object
```

In [17]:

```
#statistical view of date column
sales.date.describe()
```

Out[17]:

```
count                           9989
mean     2024-09-28 10:34:43.812193
min             2024-01-01 00:00:00
25%             2024-05-15 00:00:00
50%             2024-09-29 00:00:00
75%             2025-02-11 00:00:00
max             2025-06-30 00:00:00
Name: date, dtype: object
```

In [18]:

```
sales['date'].value_counts()
```

Out[18]:
```
date
2024-08-22    50
2024-01-06    49
2025-01-01    49
2025-06-03    49
2025-06-04    49
              ..
2025-04-09     8
2025-05-01     8
2025-06-12     8
2025-06-20     8
2025-06-23     8
Name: count, Length: 547, dtype: int64
```

In [19]:
```
sales['date'].duplicated().sum()
```

Out[19]:
```
np.int64(9442)
```

In [20]:
```
# Check sales count per day to spot any suspicious gaps
sales.groupby('date').size().sort_values()
```

Out[20]:
```
date
2025-06-12     8
2024-02-06     8
2025-02-20     8
2024-01-08     8
2025-01-15     8
              ..
2025-01-01    49
2024-01-06    49
2025-06-03    49
2025-06-04    49
2024-08-22    50
Length: 547, dtype: int64
```

In [21]:
```
# Check if any dates fall on unexpected days (e.g. if store is closed on certain days
sales['date'].dt.dayofweek.value_counts()
```

Out[21]:
```
date
5    1871
6    1833
4    1295
3    1266
0    1244
1    1244
2    1236
Name: count, dtype: int64
```

- All days have sales, but weekends (5,6) have highest volume. This makes sense for retail and validates data quality.

In [22]:
```
sales[sales['date'] == '']
```

Out[22]:

sale_id	date	store_channel	payment_method	product_id	product_name	category	sub_cat

In [23]:

```
#Analyze discount patterns
sales['discount_pct'].value_counts()
```

Out[23]:

```
discount_pct
0     4489
5     2189
10    1829
15    1127
20     355
Name: count, dtype: int64
```

- Discounts are in 5% increments (0,5,10,15,20%).
- This suggests standardized promotions rather than arbitrary discounts.
- 0% discounts (4,489 transactions) might be full-price sales or non-promotional items.

Validate and correct financial calculations

In [24]:

```
# Check if discount was calculated per unit instead of total
sales['discount_check_unit'] = (sales['unit_price'] * sales['discount_pct'] / 100).ro
sales[sales['discount_check_unit'] != sales['discount_amount']][['unit_price', 'quant
```

Out[24]:

unit_price	quantity_sold	discount_pct	discount_amount	discount_check_unit

In [25]:

```
sales['discount_pct'].describe()
```

Out[25]:

```
count    9989.000000
mean        5.329863
std         5.914879
min         0.000000
25%         0.000000
50%         5.000000
75%        10.000000
max        20.000000
Name: discount_pct, dtype: float64
```

Recalculate financial metrics

In [26]:

```
sales['discount_amount'] = (sales['unit_price'] * sales['quantity_sold'] * sales['dis
```

In [27]:

```
sales['total_sale'] = (sales['unit_price'] * sales['quantity_sold'] - sales['discount
sales['gross_profit'] = (sales['total_sale'] - sales['cost_total']).round(2)
```

In [28]:

```
# discount_amount should be unit_price * quantity * discount_pct/100
sales['discount_check'] = (sales['unit_price'] * sales['quantity_sold'] * sales['disc
sales[sales['discount_check'] != sales['discount_amount']]

# gross_profit should be total_sale - cost_total
sales['profit_check'] = (sales['total_sale'] - sales['cost_total']).round(2)
sales[sales['profit_check'] != sales['gross_profit']]
```

Out[28]:

sale_id	date	store_channel	payment_method	product_id	product_name	category	sub_cat

0 rows × 21 columns

In [29]:

```
# drop the temporary columns
sales.drop(columns=['discount_check', 'discount_check_unit', 'profit_check'], inplace
```

In [30]:

```
#check column payment_method
sales.payment_method
```

Out[30]:

```
0        Debit Card
1        Debit Card
2       Credit Card
3              Cash
4       Credit Card
           ...
9984     Debit Card
9985    Credit Card
9986    Credit Card
9987    Credit Card
9988     Debit Card
Name: payment_method, Length: 9989, dtype: object
```

In [31]:

```
#check for nique values in this column
sales.payment_method.unique()
```

Out[31]:

```
array(['Debit Card', 'Credit Card', 'Cash', 'Mobile Pay', 'PayPal', nan],
      dtype=object)
```

In [32]:

```
#Handle missing payment methods
sales.loc[sales['payment_method'].isna(), 'payment_method'] = 'Unknown'
```

In [33]:

```
#Handle missing customer_ids
sales.loc[sales['customer_id'].isna(), 'customer_id'] = 'Guest'
```

There are 101 missing values in customer_id column and could mean:

- Walk-in customers not in loyalty program.
- Online guest checkouts.

- Data entry errors.

So filling with 'Guest' category lets me analyze these separately without losing sales data.

In [34]:
```
# Verify all nulls handled
sales.isnull().sum()
```

Out[34]:
```
sale_id            0
date               0
store_channel      0
payment_method     0
product_id         0
product_name       0
category           0
sub_category       0
quantity_sold      0
unit_price         0
discount_pct       0
discount_amount    0
total_sale         0
cost_total         0
gross_profit       0
customer_id        0
is_promotion       0
is_returned        0
dtype: int64
```

In [35]:
```
#Save cleaned data
sales.to_csv('sales_clean_data.csv', index=False)
```

Data Cleaning_ customers_csv

In [36]:
```
customers.head()
```

Out[36]:

	customer_id	name	email	age	gender	city	state
0	C001	Sandra Lewis	sandra.lewis1@yahoo.com	40	F	Nashville	TN
1	C002	Kimberly Martinez	kimberly.martinez98@outlook.com	33	F	Charlotte	NC
2	C003	Susan Thompson	susan.thompson78@outlook.com	42	F	Atlanta	GA
3	C004	Anna Davis	anna.davis11@icloud.com	53	F	Minneapolis	MN
4	C005	Pamela Gonzalez	pamela.gonzalez6@yahoo.com	32	F	Phoenix	AZ

In [37]:
```
customers.tail()
```

Out[37]:

	customer_id	name	email	age	gender	city	state	seg
195	C196	Brenda Roberts	brenda.roberts78@outlook.com	39	F	Los Angeles	CA	
196	C197	Rachel Carter	rachel.carter44@outlook.com	24	F	Portland	OR	
197	C198	Raymond Nelson	raymond.nelson9@yahoo.com	36	M	Tampa	FL	
198	C199	David Mitchell	david.mitchell49@hotmail.com	35	M	Charlotte	NC	Occa
199	C200	Jason Martin	jason.martin73@icloud.com	21	M	Las Vegas	NV	

In [38]:

```
customers.dtypes
```

Out[38]:

```
customer_id                 str
name                        str
email                       str
age                       int64
gender                      str
city                        str
state                       str
segment                     str
total_items_purchased     int64
total_spent             float64
num_orders                int64
last_purchase_date          str
favourite_category          str
first_purchase_date         str
avg_order_value         float64
dtype: object
```

In [39]:

```
str_columns= ['customer_id','name', 'email','gender','city','state','segment','favour
```

In [40]:

```
customers[str_columns]= customers[str_columns].astype('object')
```

In [41]:

```
date_cols= ['last_purchase_date','first_purchase_date']
```

In [42]:

```
customers[date_cols] = customers[date_cols].apply(pd.to_datetime)
```

In [43]:

```
customers.dtypes
```

Out[43]:

```
customer_id                     object
name                            object
email                           object
age                              int64
gender                          object
city                            object
state                           object
segment                         object
total_items_purchased            int64
total_spent                    float64
num_orders                       int64
last_purchase_date       datetime64[us]
favourite_category              object
first_purchase_date      datetime64[us]
avg_order_value                float64
dtype: object
```

In [44]:

```
customers.shape
```

Out[44]:

```
(200, 15)
```

In [45]:

```
customers.describe()
```

Out[45]:

	age	total_items_purchased	total_spent	num_orders	last_purchase_date	first_p
count	200.000000	200.000000	200.000000	200.000000	200	
mean	34.800000	93.370000	3360.533800	49.945000	2025-06-19 00:28:48	
min	0.000000	47.000000	1548.280000	29.000000	2025-04-22 00:00:00	
25%	26.000000	73.000000	2712.372500	45.750000	2025-06-14 00:00:00	
50%	35.000000	86.000000	3108.060000	49.000000	2025-06-23 12:00:00	
75%	41.000000	110.500000	3939.747500	54.000000	2025-06-28 00:00:00	
max	145.000000	190.000000	6903.240000	75.000000	2025-06-30 00:00:00	
std	13.325458	27.834274	1021.145301	7.305397	NaN	

The summary statistics indicate that most numerical variables fall within reasonable and expected ranges. The only anomaly observed is in the age column, where the minimum (0) and maximum (145) values are unrealistic and require cleaning. Other metrics such as total items purchased, total spent, number of orders, and average order value show consistent distributions with no negative or extreme values.

In [46]:

```
customers[(customers['age'] < 18) | (customers['age'] > 100)][ ['customer_id', 'name'
```

Out[46]:

	customer_id	name	age
79	C080	Jacob Clark	0
116	C117	Mark Green	145

- The age column has a maximum value of 145 and a minimum value of 0, which are unrealistic and likely the result of data entry or system errors. Since these values do not represent valid customers, I will treat them as invalid observations rather than true outliers. To maintain data integrity and avoid distorting any age-based sales insights, these values will be replaced with missing (NaN) values instead of being imputed.

In [47]:

```
customers.loc[(customers['age'] < 18) | (customers['age'] > 100), 'age'] = pd.NA
```

In [48]:

```
def get_age_group(age):
    if pd.isna(age):
        return 'Unknown'
    elif age < 25:
        return '18-24'
    elif age < 35:
        return '25-34'
    elif age < 45:
        return '35-44'
    elif age < 55:
        return '45-54'
    elif age < 65:
        return '55-64'
    else:
        return '65+'

customers['age_group'] = customers['age'].apply(get_age_group)
print(customers['age_group'].value_counts())
```

```
age_group
35-44      71
25-34      57
18-24      41
45-54      22
55-64       6
Unknown     2
65+         1
Name: count, dtype: int64
```

In [49]:

```
customers.isnull().sum()
```

Out[49]:
```
customer_id              0
name                     0
email                    0
age                      2
gender                   0
city                     0
state                    0
segment                  0
total_items_purchased    0
total_spent              0
num_orders               0
last_purchase_date       0
favourite_category       0
first_purchase_date      0
avg_order_value          0
age_group                0
dtype: int64
```

In [50]:
```
customers.duplicated().sum()
```

Out[50]:
```
np.int64(0)
```

In [51]:
```
customers.gender.unique()
```

Out[51]:
```
array(['F', 'M'], dtype=object)
```

In [52]:
```
customers.to_csv('customers_clean_data.csv', index=False )
```

Data Cleaning_ products_csv

In [53]:
```
# overview of products stylesense deals with
products.head()
```

Out[53]:

	product_id	product_name	category	sub_category	base_price	cost_price	season
0	P001	Men's Basic T-shirt	Men	Shirt	19.99	7.0	All
1	P002	Men's Premium T-shirt	Men	Shirt	29.99	11.0	All
2	P003	Men's Slim Jeans	Men	Pants	49.99	18.0	All
3	P004	Men's Chino Pants	Men	Pants	44.99	15.0	All
4	P005	Men's Hoodie	Men	Jacket	59.99	22.0	Winter

In [54]:
```
#check for columns and rows
products.shape
```

Out[54]:
```
(25, 7)
```

In [55]:

```
# a quick overview of the dataset's columns and memory usage
products.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 25 entries, 0 to 24
Data columns (total 7 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   product_id    25 non-null     str
 1   product_name  25 non-null     str
 2   category      25 non-null     str
 3   sub_category  25 non-null     str
 4   base_price    25 non-null     float64
 5   cost_price    25 non-null     float64
 6   season        25 non-null     str
dtypes: float64(2), str(5)
memory usage: 1.5 KB
```

In [56]:

```
# check columns
products.columns
```

Out[56]:

```
Index(['product_id', 'product_name', 'category', 'sub_category', 'base_price',
       'cost_price', 'season'],
      dtype='str')
```

In [57]:

```
# get summary statistics of each column
products.describe()
```

Out[57]:

	base_price	cost_price
count	25.000000	25.000000
mean	40.590000	14.420000
std	20.223748	8.015194
min	9.990000	2.500000
25%	24.990000	9.000000
50%	39.990000	14.000000
75%	49.990000	18.000000
max	89.990000	35.000000

In [58]:

```
#check the datatypes
products.dtypes
```

Out[58]:
```
product_id          str
product_name        str
category            str
sub_category        str
base_price      float64
cost_price      float64
season              str
dtype: object
```

In [59]:
```
# fix the string datatypes to object
string_columns= ['product_id','product_name','category','sub_category','season']
```

In [60]:
```
products[string_columns]= products[string_columns].astype('object')
```

In [61]:
```
#check for duplicates
products.duplicated().sum()
```

Out[61]:
```
np.int64(0)
```

- There are no duplicated values.

In [62]:
```
#check for missing values
products.isnull().sum()
```

Out[62]:
```
product_id      0
product_name    0
category        0
sub_category    0
base_price      0
cost_price      0
season          0
dtype: int64
```

- No missing values.

In [63]:
```
# Verify prices are positive
print(f"All base_price > 0: {(products['base_price'] > 0).all()}")
print(f"All cost_price > 0: {(products['cost_price'] > 0).all()}")
```

```
All base_price > 0: True
All cost_price > 0: True
```

In [64]:
```
# Calculate profit metrics for profitability analysis
products['profit_per_unit'] = (products['base_price'] - products['cost_price']).round
products['profit_margin_pct'] = ((products['profit_per_unit'] / products['base_price'
```

In [65]:
```
print("\nProfit margins by category:")
print(products.groupby('category')['profit_margin_pct'].mean().round(1))
```

```
Profit margins by category:
category
All      71.6
Kids     65.4
Men      64.5
Women    64.0
Name: profit_margin_pct, dtype: float64
```

In [66]:

```python
# season flags for seasonality analysis
products['is_winter'] = products['season'].isin(['Winter', 'All'])
products['is_summer'] = products['season'].isin(['Summer', 'All'])
products['is_all_season'] = products['season'] == 'All'
```

In [67]:

```python
products.head()
```

Out[67]:

	product_id	product_name	category	sub_category	base_price	cost_price	season	profit_p
0	P001	Men's Basic T-shirt	Men	Shirt	19.99	7.0	All	
1	P002	Men's Premium T-shirt	Men	Shirt	29.99	11.0	All	
2	P003	Men's Slim Jeans	Men	Pants	49.99	18.0	All	
3	P004	Men's Chino Pants	Men	Pants	44.99	15.0	All	
4	P005	Men's Hoodie	Men	Jacket	59.99	22.0	Winter	

In [68]:

```python
products.to_csv('products_clean_data.csv', index=False)
```

Data Cleaning_ expenses_csv

In [69]:

```python
expenses.head()
```

Out[69]:

	date	category	amount	notes
0	2024-01-01	Rent	3500.00	Monthly store rent
1	2024-01-01	Insurance	400.00	Business insurance premium
2	2024-01-01	Software & Subscriptions	150.00	POS and inventory software
3	2024-01-01	Cost of Goods Sold	710.50	Daily COGS
4	2024-01-01	Marketing	723.55	Social media and promotions

In [70]:
```
expenses.shape
```

Out[70]:
```
(1322, 4)
```

In [71]:
```
expenses.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1322 entries, 0 to 1321
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   date      1322 non-null   str
 1   category  1322 non-null   str
 2   amount    1322 non-null   float64
 3   notes     1322 non-null   str
dtypes: float64(1), str(3)
memory usage: 41.4 KB
```

In [72]:
```
expenses.describe()
```

Out[72]:

	amount
count	1322.000000
mean	414.129667
std	678.576065
min	2.760000
25%	28.942500
50%	181.065000
75%	430.875000
max	3500.000000

Observations

- Skewed distribution: The max (3500) is far above the median (181), so the data is right-skewed.
- Possible outliers: High-value expenses such as those approaching $3,500, may be outliers.

In [73]:
```
expenses.dtypes
```

Out[73]:
```
date            str
category        str
amount      float64
notes           str
dtype: object
```

In [74]:
```
expenses_str_columns= ['category','notes']
```

In [75]:
```
expenses[expenses_str_columns]= expenses[expenses_str_columns].astype('object')
```

In [76]:
```
expenses['date']= pd.to_datetime(expenses['date'])
```

In [77]:
```
expenses.dtypes
```

Out[77]:
```
date        datetime64[us]
category            object
amount             float64
notes               object
dtype: object
```

In [78]:
```
expenses.isnull().sum()
```

Out[78]:
```
date        0
category    0
amount      0
notes       0
dtype: int64
```

In [79]:
```
expenses.duplicated().sum()
```

Out[79]:
```
np.int64(0)
```

In [80]:
```
# Check for non-positive expenses
neg_expenses = expenses[expenses['amount'] <= 0]
if len(neg_expenses) > 0:
    print("These might be refunds - investigate")
    print(neg_expenses[['date', 'category', 'amount', 'notes']].head())
```

- There are no negative values//expenses in the 'amount' column.

In [81]:
```
# View categories
expenses['category'].value_counts()
```

Out[81]:
```
category
Cost of Goods Sold          547
Shipping & Packaging        546
Marketing                    79
Payroll                      78
Rent                         18
Insurance                    18
Software & Subscriptions     18
Utilities                    18
Name: count, dtype: int64
```

Data Quality Summary:

- **Rows:** 1,322 (no missing values, no duplicates)
- **Date Range:** January 2024 - June 2025 (18 full months)
- **Categories:** 8 standardized expense types
- **Data Integrity:** All amounts positive, all dates valid

Outliers detection

In [82]:
```
# Calculate IQR
Q1 = expenses['amount'].quantile(0.25)
Q3 = expenses['amount'].quantile(0.75)
IQR = Q3 - Q1
```

In [83]:
```
# Define bounds
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
```

In [84]:
```
lower_bound
```

Out[84]:
```
np.float64(-573.9562500000001)
```

In [85]:
```
upper_bound
```

Out[85]:
```
np.float64(1033.77375)
```

In [86]:
```
# Flag potential outliers
outliers = expenses[(expenses['amount'] < lower_bound) |
                    (expenses['amount'] > upper_bound)]

print(f"Potential outliers: {len(outliers)}")
print(outliers[['date', 'category', 'amount', 'notes']].head())
```

```
Potential outliers: 124
         date            category  amount                     notes
0  2024-01-01                Rent  3500.0        Monthly store rent
6  2024-01-02  Cost of Goods Sold  1424.5                Daily COGS
8  2024-01-03  Cost of Goods Sold  1461.0                Daily COGS
12 2024-01-05             Payroll  2498.5  Weekly payroll - 4 staff
13 2024-01-05  Cost of Goods Sold  1223.5                Daily COGS
```

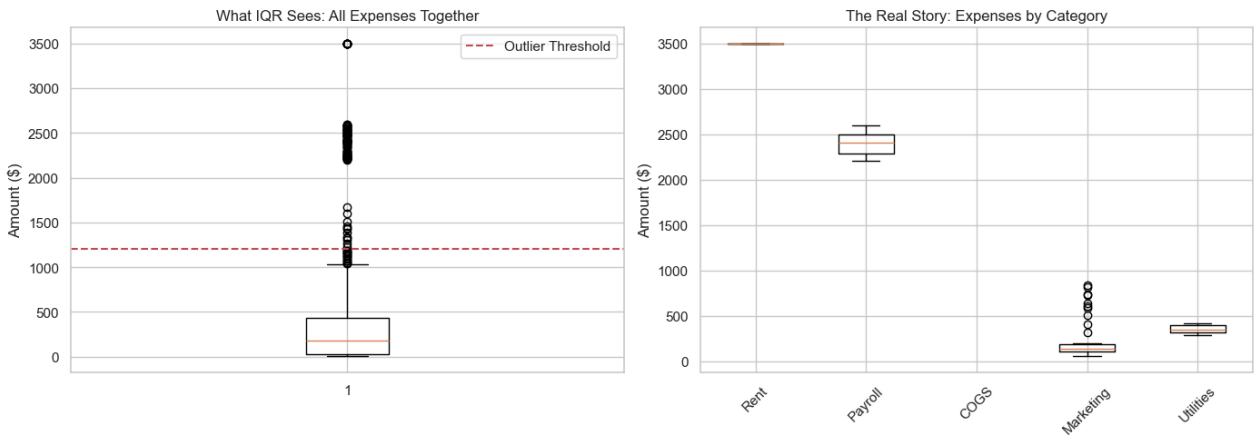
In [87]:
```
# Visualize outliers
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Plot 1: All expenses together
axes[0].boxplot(expenses['amount'])
axes[0].set_title('What IQR Sees: All Expenses Together')
axes[0].set_ylabel('Amount ($)')
axes[0].axhline(y=1200, color='r', linestyle='--', label='Outlier Threshold')
```

```
axes[0].legend()

# Plot 2: Expenses by category
categories = ['Rent', 'Payroll', 'COGS', 'Marketing', 'Utilities']
data_to_plot = [expenses[expenses['category'] == cat]['amount'] for cat in categories
axes[1].boxplot(data_to_plot, labels=categories)
axes[1].set_title('The Real Story: Expenses by Category')
axes[1].set_ylabel('Amount ($)')
axes[1].tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()
```



- The 'outliers' aren't errors, they're actually signs of healthy business activity. High COGS days represent our busiest sales periods, rent shows our stable store footprint, and payroll reflects our commitment to staffing during peak times. Removing these would hide the true story of our business performance.

Thus, i'll retain all the otlier values in the dataset for accurate analysis.

In [88]:
```
expenses.head()
```

Out[88]:

	date	category	amount	notes
0	2024-01-01	Rent	3500.00	Monthly store rent
1	2024-01-01	Insurance	400.00	Business insurance premium
2	2024-01-01	Software & Subscriptions	150.00	POS and inventory software
3	2024-01-01	Cost of Goods Sold	710.50	Daily COGS
4	2024-01-01	Marketing	723.55	Social media and promotions

In [89]:
```
expenses.to_csv('expenses_clean_data.csv', index=False)
```

Data Cleaning_inventory_csv

In [90]:
```
inventory.head()
```

Out[90]:

	date	product_id	product_name	category	stock_level	reorder_point	supplier	unit_cost
0	2024-01-01	P001	Men's Basic T-shirt	Men	200	40	Urban Threads Co.	7.0
1	2024-01-01	P002	Men's Premium T-shirt	Men	200	40	StyleForce Ltd	11.0
2	2024-01-01	P003	Men's Slim Jeans	Men	200	40	StyleForce Ltd	18.0
3	2024-01-01	P004	Men's Chino Pants	Men	200	40	Urban Threads Co.	15.0
4	2024-01-01	P005	Men's Hoodie	Men	200	40	AlphaTex Supplies	22.0

In [91]:
```
inventory.shape
```

Out[91]:
```
(13675, 9)
```

In [92]:
```
inventory.columns
```

Out[92]:
```
Index(['date', 'product_id', 'product_name', 'category', 'stock_level',
       'reorder_point', 'supplier', 'unit_cost', 'stock_value'],
      dtype='str')
```

In [93]:
```
inventory.duplicated().sum()
```

Out[93]:
```
np.int64(0)
```

In [94]:
```
inventory.isnull().sum()
```

Out[94]:
```
date              0
product_id        0
product_name      0
category          0
stock_level       0
reorder_point     0
supplier          0
unit_cost         0
stock_value      15
dtype: int64
```

In [95]:
```
# Fill missing stock_value with calculated values
inventory['stock_value'] = inventory['stock_value'].fillna(
```

```
    inventory['stock_level'] * inventory['unit_cost']
)
```

In [96]:

```
inventory.isnull().sum()
```

Out[96]:

```
date             0
product_id       0
product_name     0
category         0
stock_level      0
reorder_point    0
supplier         0
unit_cost        0
stock_value      0
dtype: int64
```

In [97]:

```
inventory.dtypes
```

Out[97]:

```
date                 str
product_id           str
product_name         str
category             str
stock_level        int64
reorder_point      int64
supplier             str
unit_cost        float64
stock_value      float64
dtype: object
```

In [98]:

```
inv_str_cols= ['product_id','product_name','category','supplier']
```

In [99]:

```
inventory[inv_str_cols] = inventory[inv_str_cols].astype('object')
```

In [100…

```
inventory['date']= pd.to_datetime(inventory['date'])
```

In [101…

```
inventory.dtypes
```

Out[101…

```
date             datetime64[us]
product_id               object
product_name             object
category                 object
stock_level               int64
reorder_point             int64
supplier                 object
unit_cost               float64
stock_value             float64
dtype: object
```

In [102…

```
# Check for negative values in numeric columns
negatives = inventory[(inventory['stock_level'] < 0) |
                      (inventory['unit_cost'] < 0) |
                      (inventory['stock_value'] < 0)]
```

```
print(f"\nNegative values found: {len(negatives)}")
if len(negatives) > 0:
    print(negatives)
```

```
Negative values found: 0
```

In [103…

```
# Validate 'stock_value' consistency
inventory['calculated_stock_value'] = inventory['stock_level'] * inventory['unit_cost
mismatched = inventory[inventory['stock_value'] != inventory['calculated_stock_value'
print(f"\nMismatched stock values: {len(mismatched)}")
if len(mismatched) > 0:
    print(mismatched[['product_id','stock_value','calculated_stock_value']])
```

```
Mismatched stock values: 0
```

In [104…

```
print(inventory['category'].unique())
```

```
['Men' 'Women' 'Kids' 'All']
```

In [105…

```
print(inventory['supplier'].unique())
```

```
['Urban Threads Co.' 'StyleForce Ltd' 'AlphaTex Supplies'
 'Bella Fabrics Ltd' 'TrendSet Suppliers' 'KidWear Distributors'
 'TinyTrend Co.' 'GlobalGear Wholesale']
```

In [106…

```
inventory = inventory.sort_values(['date', 'product_id']).reset_index(drop=True)
```

In [107…

```
inventory= inventory.drop(['calculated_stock_value'], axis=1)
```

In [108…

```
inventory.describe()
```

Out[108…

	date	stock_level	reorder_point	unit_cost	stock_value
count	13675	13675.000000	13675.0	13675.000000	13675.000000
mean	2024-09-30 00:00:00	105.014845	40.0	14.420000	1509.847057
min	2024-01-01 00:00:00	40.000000	40.0	2.500000	100.000000
25%	2024-05-16 00:00:00	71.000000	40.0	9.000000	748.000000
50%	2024-09-30 00:00:00	102.000000	40.0	14.000000	1320.000000
75%	2025-02-14 00:00:00	134.000000	40.0	18.000000	2052.000000
max	2025-06-30 00:00:00	200.000000	40.0	35.000000	7000.000000
std	NaN	40.012472	0.0	7.853541	1006.661812

In [109…

```
inventory.stock_level.unique()
```

Out[109…

```
array([200, 199, 196, 198, 195, 197, 193, 184, 192, 191, 194, 182, 189,
       186, 187, 180, 190, 177, 176, 188, 185, 169, 179, 173, 174, 175,
       161, 178, 165, 172, 168, 150, 158, 171, 181, 166, 167, 138, 159,
       157, 164, 136, 162, 135, 156, 170, 155, 163, 134, 152, 132, 154,
       149, 160, 131, 153, 146, 124, 148, 142, 151, 122, 147, 139, 118,
       145, 116, 114, 141, 137, 111, 143, 144, 101, 133,  97, 128, 140,
        93, 130,  91,  90, 126,  87, 123, 129,  85, 121, 183,  79, 117,
       119,  74, 127, 125,  73, 113,  71, 110, 120, 107,  68, 104,  66,
       103, 115,  99,  65, 108,  95,  61,  92, 102,  94, 105,  56,  89,
       100, 112,  50,  88,  43,  86,  84, 106, 109,  77,  80,  83,  72,
        78,  98,  63,  67,  75,  96,  69,  64,  62,  57,  55,  82,  59,
        76,  53,  58,  81,  51,  52,  49,  48,  54,  60,  46,  45,  42,
        41,  70,  40,  47,  44])
```

In [110…

```python
# Select specific columns
inventory_view = inventory[['date', 'product_name', 'stock_level', 'reorder_point']]

# Display the result
print(inventory_view)
```

```
           date           product_name  stock_level  reorder_point
0     2024-01-01    Men's Basic T-shirt          200             40
1     2024-01-01  Men's Premium T-shirt          200             40
2     2024-01-01       Men's Slim Jeans          200             40
3     2024-01-01      Men's Chino Pants          200             40
4     2024-01-01           Men's Hoodie          200             40
...          ...                     ...          ...            ...
13670 2025-06-30  Unisex Socks (3-pack)           43             40
13671 2025-06-30           Leather Belt           42             40
13672 2025-06-30        Canvas Tote Bag           62             40
13673 2025-06-30      Men's Swim Shorts           60             40
13674 2025-06-30       Women's Swimsuit          115             40

[13675 rows x 4 columns]
```

In [ ]:

```
Handling
```

In [111…

```python
# Calculate day-over-day change
inventory = inventory.sort_values(['product_id', 'date'])
inventory['stock_change'] = inventory.groupby('product_id')['stock_level'].diff()
```

In [112…

```python
# Flag large increases (> 50 units)
inventory['is_suspect'] = inventory['stock_change'] > 50
print("Suspicious stock increases found:", inventory['is_suspect'].sum())
```

```
Suspicious stock increases found: 136
```

In [113…

```python
# Estimate daily sales (stock decreases only)
inventory['daily_sales'] = inventory['stock_change'] * -1
```

In [114…

```python
# Remove negative sales (these are stock increases)
inventory['daily_sales'] = inventory['daily_sales'].clip(lower=0)
```

In [115…

```
# Ignore suspicious large increases
inventory.loc[inventory['is_suspect'], 'daily_sales'] = 0
```

In [116…

```
# Example metric
avg_sales = inventory.groupby('product_name')['daily_sales'].mean()

print("\nTop 5 Products by Average Daily Sales:")
print(avg_sales.sort_values(ascending=False).head())
```

```
Top 5 Products by Average Daily Sales:
product_name
Canvas Tote Bag          2.624542
Unisex Cap               2.505495
Leather Belt             2.369963
Unisex Socks (3-pack)    2.256410
Men's Chino Pants        1.463370
Name: daily_sales, dtype: float64
```

In [117…

```
inventory.head()
```

Out[117…

	date	product_id	product_name	category	stock_level	reorder_point	supplier	unit_cos
0	2024-01-01	P001	Men's Basic T-shirt	Men	200	40	Urban Threads Co.	7.
25	2024-01-02	P001	Men's Basic T-shirt	Men	199	40	Urban Threads Co.	7.
50	2024-01-03	P001	Men's Basic T-shirt	Men	195	40	Urban Threads Co.	7.
75	2024-01-04	P001	Men's Basic T-shirt	Men	195	40	Urban Threads Co.	7.
100	2024-01-05	P001	Men's Basic T-shirt	Men	195	40	Urban Threads Co.	7.

In [119…

```
# Replace NaN sales with 0
inventory['daily_sales'] = inventory['daily_sales'].fillna(0)

# Convert -0.0 to 0
inventory['daily_sales'] = inventory['daily_sales'].astype(int)
```

In [120…

```
inventory.head()
```

Out[120…

	date	product_id	product_name	category	stock_level	reorder_point	supplier	unit_cos
0	2024-01-01	P001	Men's Basic T-shirt	Men	200	40	Urban Threads Co.	7.
25	2024-01-02	P001	Men's Basic T-shirt	Men	199	40	Urban Threads Co.	7.
50	2024-01-03	P001	Men's Basic T-shirt	Men	195	40	Urban Threads Co.	7.
75	2024-01-04	P001	Men's Basic T-shirt	Men	195	40	Urban Threads Co.	7.
100	2024-01-05	P001	Men's Basic T-shirt	Men	195	40	Urban Threads Co.	7.

In [121…

```
inventory.to_csv('inventory_clean_data.csv', index=False)
```

In [ ]: